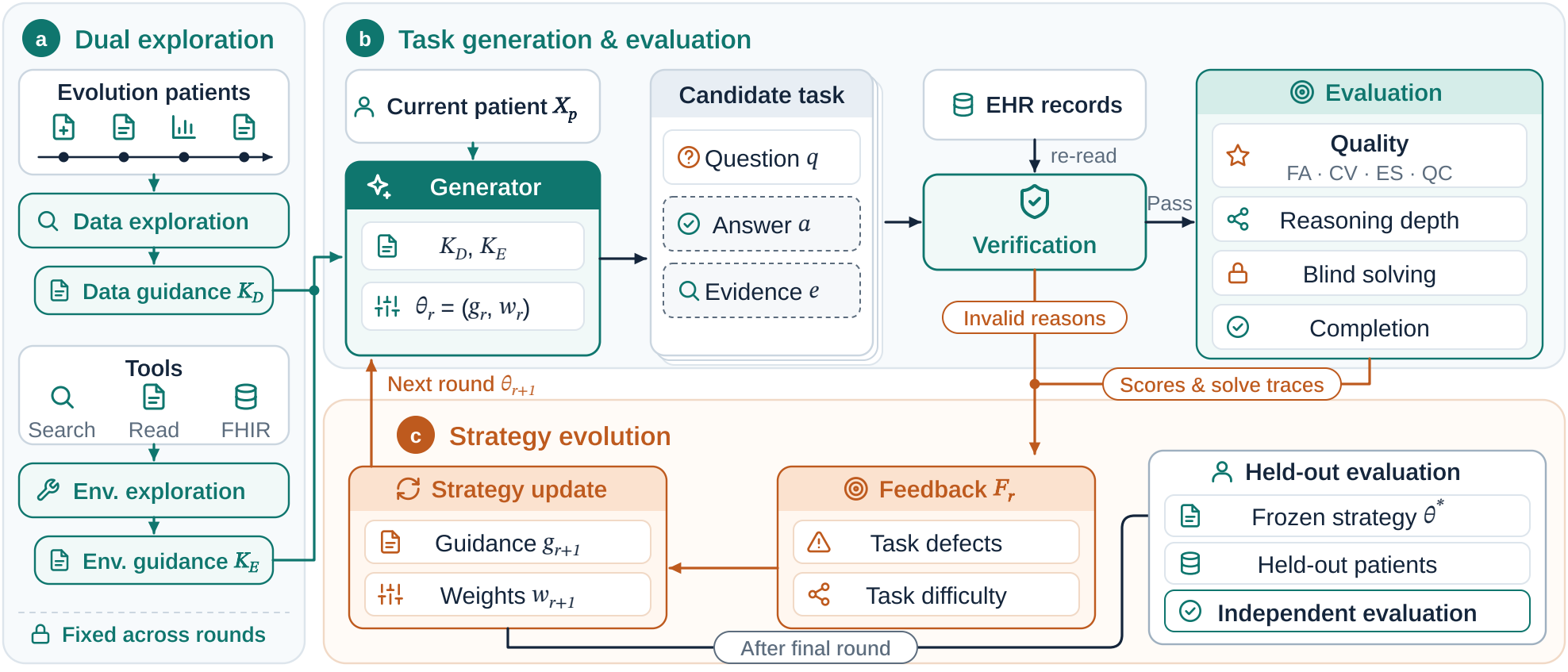

a Dual exploration
Evolution patients
Data exploration
Data guidance K_D
Tools
Search
Read
FHIR
Env. exploration
Env. guidance K_E
Fixed across rounds
b Task generation & evaluation
Current patient X_p
Generator
K_D, K_E
$\theta_r = (g_r, w_r)$
Candidate task
Question q
Answer a
Evidence e
EHR records
re-read
Verification
Pass
Invalid reasons
Evaluation
Quality
FA · CV · ES · QC
Reasoning depth
Blind solving
Completion
Scores & solve traces
Next round θ_{r+1}
c Strategy evolution
Strategy update
Guidance g_{r+1}
Weights w_{r+1}
Feedback F_r
Task defects
Task difficulty
After final round
Held-out evaluation
Frozen strategy θ^*
Held-out patients
Independent evaluation